United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam   Trimester: Fall 2024   Marks: 40   Time: 2 Hours
Code: CSE 3411   Course Title: System Analysis & Design

"Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules."

Answer the following questions

Question 1: [CO3] [15]

Consider the following scenario:

The university is introducing a smart bicycle-sharing system called "CampusRide" to promote sustainable and convenient travel across campus. The system provides users with a network of smart bicycles parked at designated stations. Users can interact with the system through a mobile app, which allows them to locate available bicycles, reserve one, unlock it, and track their rides.

The system supports three types of users:
1. Students, and university employees are the primary users. They can locate bicycles, reserve them, track their usage statistics (e.g., distance traveled, calories burned), and report issues like damaged bicycles. They also receive notifications for nearby stations with available bikes and reminders to return bicycles within the allowed usage period.
2. Technicians handle maintenance tasks. They monitor bicycles for reported issues, retrieve damaged ones for repair, and ensure stations are stocked with enough bicycles.
3. Administrators oversee the CampusRide system. They manage the bicycle inventory, monitor station activity, and generate detailed reports on system usage and performance. Administrators can also lock bicycles or stations temporarily for maintenance or during emergencies.

The system is integrated with the university's identity management system for authentication and supports payment for extended usage through the app. All activities, such as reservations, usage, and maintenance logs, are tracked for reporting and security purposes.

a. Design a Use Case Diagram for the "CampusRide" application. Ensure you include multiple actors (students, faculty, technicians, administrators, etc.) and demonstrate at least one **include** or **extend** relationship. **[3]**

b. Draw the Class Diagram and CRC cards for the "CampusRide" application, identifying the main classes, their attributes, methods, and relationships. Make sure to include classes like **User, Bicycle, Station,** and **RideHistory.** **[4+2]**

c. Create a State Diagram that depicts the lifecycle of a bicycle in the "CampusRide" system, from being available at a station to reserved, in use, returned, and potentially under maintenance. **[3]**

d. Draw a Sequence Diagram illustrating the process of a user locating a bicycle, reserving it through the app, unlocking it, and reporting an issue after completing the ride. **[3]**

Question 2: [CO4] [5]

a. Imagine you and your team are planning to launch a solar power plant as part of a renewable energy initiative. The project requires an initial investment of $25,000, which will cover the costs of solar panels and basic infrastructure. You had to bear additional costs for maintenance and extra solar panels. The additional costs are: $4000 and $6000 in years 2 and 4 respectively.

   The projected revenue from selling electricity is $3000, $6000, $11000, $15000, and $20000 in the years 1 through 5.

   Considering yearly interest rate of 10%, analyze whether this solar power project is financially viable using the **Net Present Value (NPV)** method. **[3]**

b. Mention the concept of **Economic/Financial feasibility**. Why it is important to analyze before an IT product development. **[2]**

Question 3: [CO3] [15]

a. Create the proposed UI designs for the pages that allow users to **locate available bicycles** and **track their rides** based on the project described in **Question 1**. Ensure the designs are user-friendly, intuitive, and visually appealing. **[3+3]**
b. Which **Golden Rules of UI Design** did you consider while designing the two pages in 3(a)? **[3]**
c. Briefly explain how an effective **Software Requirement Specification (SRS)** ensures the three fundamental **3C properties**: Completeness, Correctness, and Consistency. **[3]**
d. Identify the **functional** and **non-functional** requirements for the scenario described in Question 1. **[3]**

Question 4 : [CO3] [5]

Write down an SRS document and present it as a group having the following parts on your software project work covering: Introduction, Problem Statement, Information Gathering and System Study, System Analysis, Feature list fixation with Functional and Non-functional requirements, Different Feasibility Analysis, System design (UML diagrams), UI design through Figma software, implementation plan etc. (**no need to answer this question in the examination**, it has already been evaluated in classroom activities).